



The role of fairness norms the household-based natural forest conservation: The case of Wolong, China

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ABSTRACT

The shadow of “the tragedy of the commons” with its popularized assumption of selfish individual behavior concerns policy-makers, and consequently the cooperative capability of local communities in common resource management is usually underrated. Nevertheless, here we propose a hypothesis, based on a game theory model, that the social norm of fairness rather than self-interest might motivate cooperation in natural resource conservation by discounting the utility gain from illegal logging at the cost of another household's subsidy loss. In Wolong Nature Reserve of China, a recent household-based natural forest conservation program has led to remarkable progress in protecting the habitat of giant pandas (*Ailuropoda melanoleuca*) with low monitoring pressure. We undertook experiments based on a one-shot Ultimatum Game to test the prevalence of the norm of fairness in the local communities. Most proposers in the experiments showed strong preferences for fair offers in spite of the responders' reluctance to reject low offers, and the result contradicted the expectation of pure self-interest. Taking into account the norm of fairness, the prediction of the model is consistent with the local performance of the household-based conservation policy. It highlights the potential of local social norms to facilitate participation and cooperation by the local community in common natural resource conservation.

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1. Introduction

The model of *Homo economicus*, which has guided the mainstream neoclassical economics for a long time, assumes that people are narrowly self-regarding egoists (Gintis, 2000). In accordance with this view, the tragedy of the commons seems unavoidable as everyone struggles for his or her own interests and loses the game as a whole (Hardin, 1968). However, empirical case studies (Ostrom, 1998; Ostrom et al., 2002; Dietz et al., 2003) have repeatedly rejected the assumption of universal selfishness by suggesting that people have the capability to pro-socially cooperate with each other, thus giving the common goods game a happy ending. In these cases, members of a group usually conform to some social norms, i.e., informal public rules that shape the social behavior in a certain regular and observable pattern (Bicchieri, 2005). For instance, empirical experiments have shown that the social norm of fairness is widely held in human societies (Sanfey et al., 2003; Fehr and Fischbacher, 2004; Bicchieri, 2005; Henrich et al., 2005; Henrich et al., 2006; Dawes et al., 2007). When pro-social norms are strong enough to override self-interested motives, cooperation in common resource management could be self-enforced. As a result, social capital provides a possible route for sustainable management and

governance of common natural resources (Pretty, 2003), and bounded rationality models might be of greater use in environmental research and policymaking (Venkatachalam, 2008).

In China, the central government has recently launched a national Natural Forest Conservation Program (NFCP) as a critical project for preserving ecosystem services (Wang et al., 2007). A household-based management policy has been implemented in some protected areas, where each household or group of households is assigned a patch of natural forest to monitor and receives a certain annual subsidy according to the contract if no illegal logging occurs within the patch. Local governments often intuitively assume that the material subsidy motivates the local households to enforce the forest monitoring, but the subsidy budget for conservation is usually insufficient to cover the cost of intensive monitoring (Liu et al., 2003). Nevertheless, some cases shed light on the deadlock. For instance, the household-based NFCP in Wolong Nature Reserve (Fig. 1), one of the largest giant panda (*Ailuropoda melanoleuca*) reserves, has so far worked effectively by improving conservation of the forests and panda habitat while keeping the monitoring intensity low. The evidence suggests that there could be something else at work other than the material incentives to enforce social cooperation in forest conservation (Ostrom, 2005). One of the potential factors that need to be studied is the social norm of fairness.

The objective of this study is to unveil the possible role the social norm of fairness could play in common natural resource conservation. We first modeled the household behavior in the NFCP to analyze the potential impact of the fairness norm on the equilibria and dynamics of

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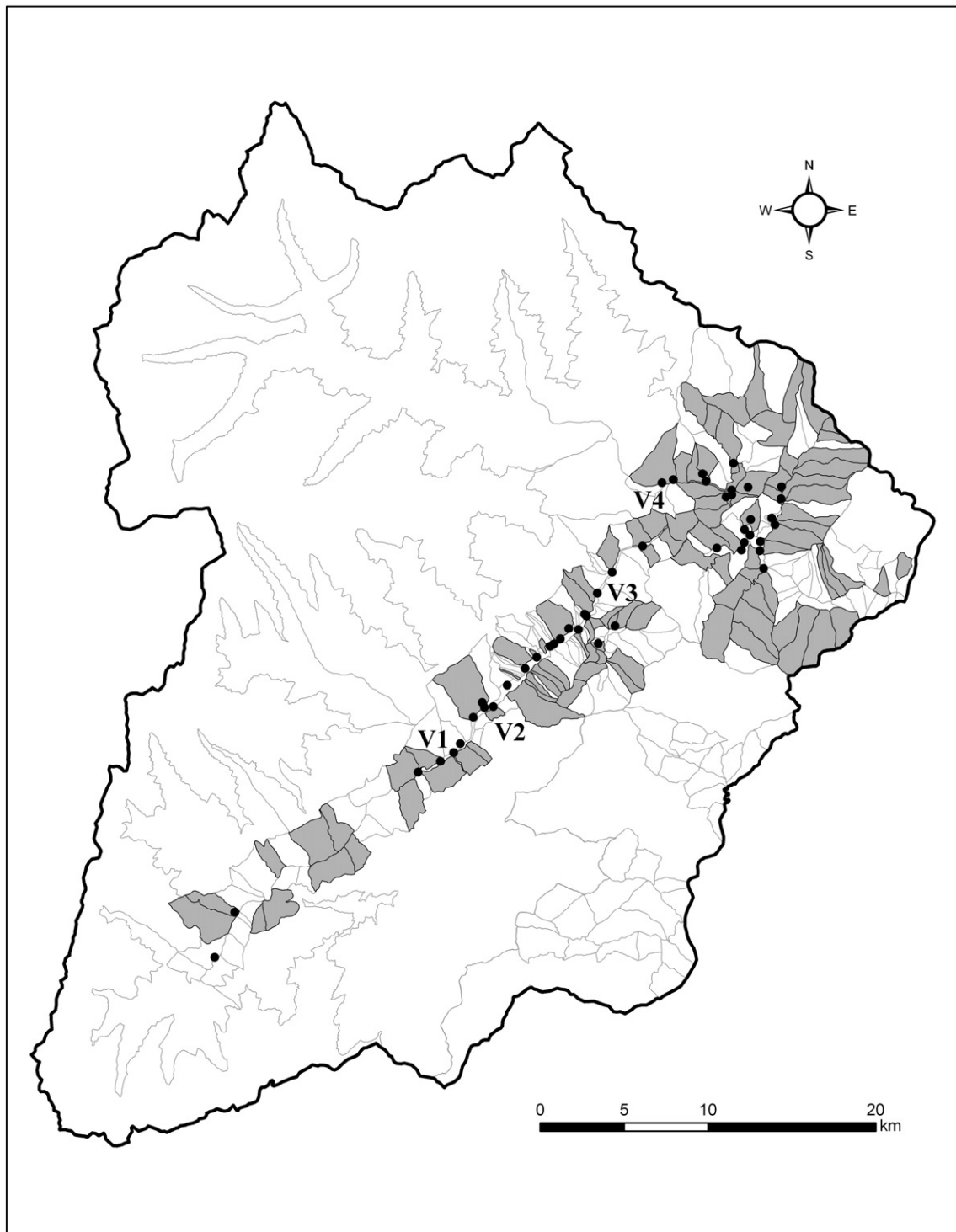


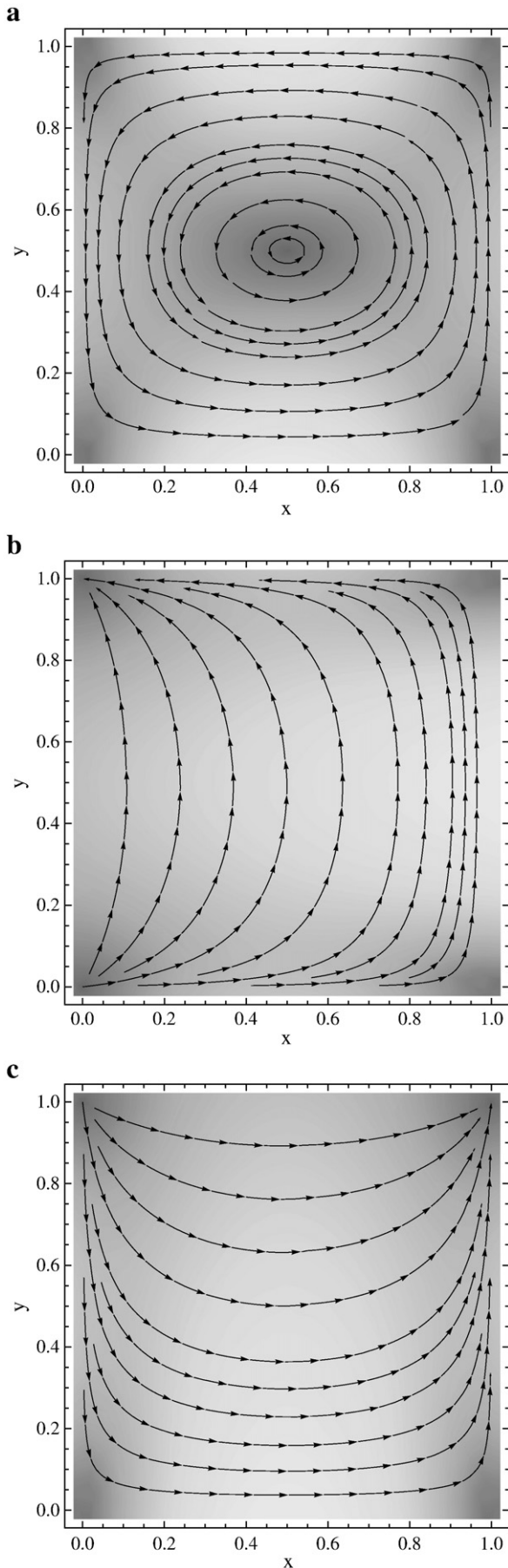
Fig. 1. The map of Wolong Nature Reserve. The gray color marks household-monitored natural forest patches. Black dots mark the distribution of residences, and experiments were undertaken in: V1 – Wolongguan, V2 – Huahongshu, V3 – Yusidong, and V4 – Gengda.

the system. Second, we undertook experiments based on the one-shot Ultimatum Game (UG) in Wolong to test how the local residents actually conform to the social norm of fairness in a general context. Based on the model and the experiment data, we made a prediction about the possible impact of the fairness norm on cooperation in the household-based NFCP and compared it with the data from a field survey. Moreover, we discussed the evolution and variation of the social

norm in the changing environment, and the possible consequences for the conservation of common natural resources.

2. The Household-based NFCP Model

In the model of the household-based NFCP, each household or group of households assigned a patch of natural forest acts as a player



that is able to make decisions independently, and each player plays two roles simultaneously. For the first role, each one plays either as a cooperator (C, who doesn't log in others' patches) or as a defector (D, who logs in others' patches). For the second role, each one chooses to monitor the forest (M) or not to monitor (N). We assume there is no correlation between the two roles of a specific player.

This is a population game with infinite rounds. In each round, b denotes the net material benefit from illegal logging, S is the subsidy per round, c is the material loss for failing to monitor one's own forest according to the contract, e is the monitoring effort, and p is the punishment against the defector if he or she is prosecuted. We assume the inequality $b < c$ holds so that a player won't become better off by logging in his or her own patch.

Thus we obtain the payoff matrix of a stage game, U_b being the utility gain of a defector from logging and U_c the utility loss suffered by those who don't monitor due to logging.

	M	N
C	$0, S - e$	$0, S$
D	$-p, S - e$	$U_b, S - U_c$

Here we take into account not only self-interested incentives but also the regard for fairness in the linear utility functions, as follows (Fehr and Schmidt, 1999):

$$\begin{aligned} U_c &= c + \alpha(b + c) \\ U_b &= b - \beta(b + c). \end{aligned} \quad (1)$$

The parameters α and β respectively represent the sensitivity to the negative feelings borne by a defector (e.g., guilt or anxiety) and by one who doesn't monitor (e.g., envy or anger) when they interact with each other with the norm of fairness violated. A purely self-interested player, for whom $\alpha = \beta = 0$, doesn't care about fairness. For simplicity, we do not distinguish the motive of inequity aversion from other related motives like guilt aversion.

We first assume a population that has homogeneous conformity to the norm of fairness and is well mixed with equal probability of players randomly meeting each other. Each player updates his or her strategy using a proportional imitation rule such that those who are relatively worse off in a given round tend to try the other strategy in the next round, with the probability of switching in proportion to the current payoff disadvantage (Hofbauer and Sigmund, 1998; Schlag, 1998). Let x_1, x_2, y_1 and y_2 denote the proportions of cooperators, defectors, non-monitors and monitors in the population, respectively, so $x_1 + x_2 = y_1 + y_2 = 1$. Defining f_i and g_j as the payoffs of aggressive strategy i and defensive strategy j , f and g the corresponding average payoffs, and v and w the respective intensities of selection, we obtain the replicator equations:

$$\begin{aligned} \dot{x}_i &= vx_i(f_i - \bar{f}) \\ \dot{y}_i &= wy_i(g_i - \bar{g}). \end{aligned} \quad (2)$$

Letting $x = x_1, y = y_1$, we then obtain from Eqs. (1) and (2) the replicator equations

$$\begin{aligned} \dot{x} &= vx(1-x) \left[1 - \left(1 + \frac{U_b}{p} \right) y \right] \\ \dot{y} &= wy(1-y) \left(\frac{e}{U_c} - 1 + x \right). \end{aligned} \quad (3)$$

Fig. 2. The population dynamics of strategies in the NCFP. Arrows indicate directions. (a) $U_b > 0, U_c > e$. The trajectory is constant circles around the unstable equilibrium $(1 - e/U_c, p/(p + U_b))$. (b) $U_b > 0, e > U_c$. The equilibrium $(0, 1)$ is globally stable and the final population is composed of pure defectors. (c) $U_b < 0$. The equilibrium $(1, 1)$ is globally stable and the final population is composed of pure cooperators.

From the replicator Eq. (3), it is obvious that $(0, 0)$, $(1, 0)$, $(0, 1)$, $(1, 1)$ and $(1 - e/U_c, p/(p + U_b))$ are all equilibria, but the dynamics depend on the values of the parameters. Given the default parameter settings, $U_b > 0$, $U_c > e$, no pure strategy is dominant, and the trajectory appears as a series of constant cycles around the unstable interior equilibrium $(1 - e/U_c, p/(p + U_b))$ (Fig. 2a). If the needed monitoring effort input, e , exceeds U_c , while U_b remains positive, not to monitor will be a dominant defensive strategy, and all the trajectories will converge to the stable point $(0, 1)$ without cooperators (Fig. 2b).

An optimal stable equilibrium exists only on the condition that $U_b < 0$, i.e., $\beta > \beta_0$, where the threshold $\beta_0 = b/(b + c) \geq b/(b + S)$. In other words, when the material benefit from logging in others' patches could not offset the psychological suffering after violating the fairness norm, cooperation is a dominant strategy, and the equilibrium is asymptotically stable (Fig. 2c).

Furthermore, if conformity to the norm among the population is heterogeneous and the proportion of the players is q for whom $\beta > \beta_0$ holds, there are always at least, but not limited to, a fraction of q cooperators. This indicates that, given the economic variables b , c , e , and S (especially when the material subsidy S is limited and comparable to the monitoring effort e) conformity to the fairness norm – particularly a negative feeling upon violating the social norm – could play a big role in motivating cooperation for conservation.

3. The Case Study and Experimental Design

3.1. The Case Study Background

The Wolong Nature Reserve was designated in 1963 for protecting the endangered giant panda, and was expanded to approximately 2000 km² in 1975. Natural forests and meadows together cover about 98% of the area in the reserve, providing habitats for over 4000 plant species, 1700 invertebrate species and 450 vertebrate species (Wolong Nature Reserve, 2005).

A population of about 5000 residents, over 80% of whom are Tibetans, now live in six villages in the reserve. The traditional pastoralists have changed into sedentary farmers after the introduction of lowland crops such as maize, potato and cabbage, and stall-fed livestock like pigs. In recent years, growing numbers of small households and shrinking farmlands, together with intensified interactions with external markets, have forced the local people to seek new employment for cash income. However, education and information deficits discourage them from competing for urban jobs. Temporary labor work in local infrastructure construction was an important but unstable form of employment, and the daily wage was around 50 RMB (about 7 US dollars) in 2007. A small fraction of the local households deal in business related to tourism.

For everyday living needs (cooking and heating), especially in the half-year-long cold winter, fuelwood collection was identified as one of the most severe anthropogenic threats to panda habitat (An et al., 2001; Liu et al., 2001). The consequent degradation of forests continued even after the reserve was established and deforestation was prohibited. The reserve administration forced conservation by official monitoring and sanctioning, but the policy not only proved to be inefficient but also repressed any cooperative incentives of the local residents. The recent availability of electricity as an alternative energy partly alleviated the conflict, but it could not replace fuelwood as the primary energy source because of the residents' limited ability to afford the electric bill (An et al., 2002).

It was after the implementation of the household-based NFPC (Wang et al., 2007) in Wolong that forest cover began to be restored in the reserve. The forests on both sides of the valley in which the residents live were divided into patches and assigned to local households to monitor. Each household received an annual subsidy of around 900 RMB per household, on average, if no illegal logging occurred within their specific patch. The residents have no property right to consume or allocate the resource in the forest. Another smaller project, the Grain for Green Program (Xu et al., 2006), was launched almost simultaneously to

reforest the sloping farmland to control erosion, but it has a relatively limited direct impact on the natural forest conservation. It was reported in 2003 that the household-level fuelwood consumption had decreased by at least 40% since NFPC was implemented in the reserve, and a further decrease of wood logging per capita from 1.4 m³ to 0.3 m³ was recorded (Wolong Nature Reserve, 2005; Viña et al., 2007). According to a field survey in 2005, only 5 sites out of 387 were found with illegal logging trails (Liu et al., unpublished data, 2005). But surprisingly, the residents actually patrol at a very low frequency, commonly once per month or so, suggesting that it is inappropriate to attribute the enhanced community cooperation in forest conservation to the monitoring intensity.

3.2. Experimental Design

We undertook the Ultimatum Game (UG) based experiments (Camerer, 2003) in four villages in the reserve (Fig. 1), including three in Wolong Township, namely Wolongguan, Huahongshu and Yusidong (V1, V2, V3 for short), and Yicun (V4) in Gengda Township. Subjects were voluntarily recruited from 31 households on average in each village, 54% of whom were female and 89.4% were Tibetans (Table 1).

The parsimonious context-free design of our experiment referred to a cross-cultural experiment program (Henrich et al., 2006). We did not mention anything about NFPC in the experiment to minimize the potential experimenter demand effects (Zizzo, 2008). At the beginning of each game we explained to the subject the rules of the game, and then tested whether he or she fully understood. Only those who understood the rules were eligible for the experiment. We randomly paired the subjects in a game and kept them anonymous to each other.

The first player, the 'proposer', was provisionally allotted a sum of cash worth 50 RMB (about 7 US dollars). We then asked the proposer to make a decision to share any portion of the money (in the unit of 5 RMB) with a second person, the 'responder'. Before being informed of the actual offer, the responder was asked to decide whether to accept each possible offer from 0 to 50 RMB. If the actual amount offered was accepted, then the responder received the amount offered and the proposer received the remainder. If the actual offer was rejected, then both subjects received nothing. In such an anonymous one-shot game, if the subjects are rational and only self-interested, the expected equilibrium is that the proposer should make the lowest offer (5 RMB) while the responder should accept any positive offer because something is better than nothing. However, if there are other-regarding subjects who care about fairness, deviations from the equilibrium are expected in the game.

In V1, V2 and V3, after allotting a proposer the stake, we asked him or her one more question to imitate a Dictator Game (DG) before the UG was introduced, leaving all the subsequent steps unchanged: How much would you like to offer if the responder is not given an option to reject the offer? The expected answer is 0 for a purely self-interested proposer whether he or she is risk-averse or not, because there is no risk of being rejected in a DG.

We gathered subjects together in V1 and V4, and undertook another two sessions door-to-door in V2 and V3, so that it would be possible to examine the unknown impact from potential communications and collusion between subjects. Before each experiment, we took household surveys to collect demographic information on the households, and we also had post-game one-to-one interviews with some subjects to learn their personal perceptions of the experiment and the motivations behind their choices.

Table 1

The demographic data of the subjects in the experiments.

Age (year)	Male	Tibetan ^a	Education (year)	Family size	Head of family
43.2 ± 1.2	45.50%	89.4%	5.1 ± 0.3	5.3 ± 0.2	46.30%

^a Other ethnic groups are Han and Qiang.

4. Results

4.1. The Experimental Results

The results of the experiments are presented in Table 2. The mean offer of the Ultimatum Game is 50.6% ($\pm 0.9\%$, s.e.m.), showing a strong deviation from the expectation of self-interest maximization (Wilcoxon rank sum test, $p < 0.0001$). The unique modal offer of the games in the four villages is the fair offer, 50%, which takes a portion of 85.9% in the total number of the offers. Sub-fair offers are rare, taking only 2 out of 64, while all the rest 11% are hyper-fair offers (Fig. 3a).

The responders' minimum acceptable offers (MAO) show a bimodal distribution, with 0 as the primary modal MAO and 50% as the sub-modal MAO (Fig. 3b). The MAO is $17.0\% \pm 2.6\%$, which is strongly lower than the mean offer (bootstrap test, $p < 0.0001$). There was no sub-fair offer actually rejected in the games. Given the actual distribution of MAO, we calculated the expected income of the proposers with different offers, and the result shows that the minimum positive offer, 10% (5 RMB), is the optimal option that would lead to the highest expected income (Fig. 4), and 20% (10 RMB) follows. This prediction is strongly different from the actual offer distribution (Wilcoxon rank sum test, $p < 0.0001$), so income-maximizing incentives cannot explain the behavior of the proposers.

Risk-aversion is another possible explanation to be tested, and we could not exclude the possibility based on the limited information solely from the UG result. However, the answers to the DG question show a consistent pattern ($51\% \pm 0.8\%$) with the UG offers (Fig. 5), with no difference apparent in the pairwise test (Wilcoxon rank sum test, $p = 0.8086$). As a result, it indicates that risk-aversion is not a sound explanation for interpreting the preference for fair offers, either.

Another result worth noting is the strongly lower maximum acceptable offer (MXAO) compared with the rational expectation (Mean = 70.4%, Schlag mean test (Schlag, 2008), $p < 0.001$). The distribution is bimodal with the two peaks at 50% (the relative frequency is 43.9%) and 100% (the relative frequency is 29.8%), which indicates the striking fact that a large fraction of subjects did not want to accept the hyper-fair offer. Furthermore, a test comparing the deviations from fair offer between MAO and MXAO shows that the responders showed an unusual stronger sensitivity against the positive deviations (Wilcoxon rank sum test, $p = 0.0001$).

Furthermore, the differences in the experiment results across villages are not significant (Table 3), indicating that patterns of conformity to the social norm of fairness were consistent among the local communities and that the different experimental methods had no significant impact on the results.

Table 2
Results of the Ultimatum Game experiments.

Village	V1 ^a	V2 ^b	V3 ^c	V4 ^d	Total
No. pairs	17	15	16	16	64
Stake (RMB)	50	50	50	50	50
Offer					
Mean %	48.8	50.6	51.8	51.2	50.6
Model %	50	50	50	50	50
SD %	12.6	2.6	4.0	3.4	7.0
MAO					
Mean %	12.4	23.4	14.6	17.9	17.0
Model %	0	10	0	0	0
SD %	19.2	23.2	19.4	22.6	21.0
MXAO					
Mean %	74.8	72.6	60.8	71.4	70.4
Model %	100	50/100	50	50	50
SD %	21.8	23.8	19.0	21.4	21.8

^a Wolongguan.

^b Huahongshu.

^c Yusidong.

^d Gengda.

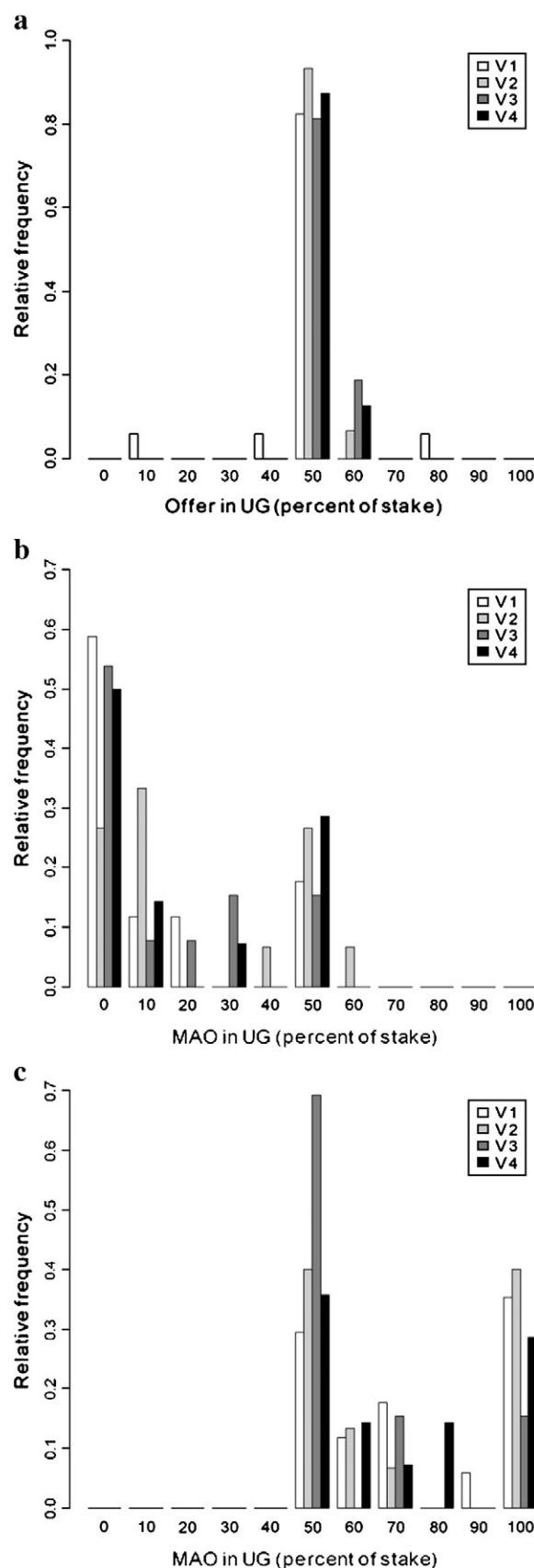


Fig. 3. The results of the Ultimatum Games. (a) Offers (in percentage); (b) minimum acceptable offers (MAO, in percentage); and (c) maximum acceptable offers (MXAO, in percentage).

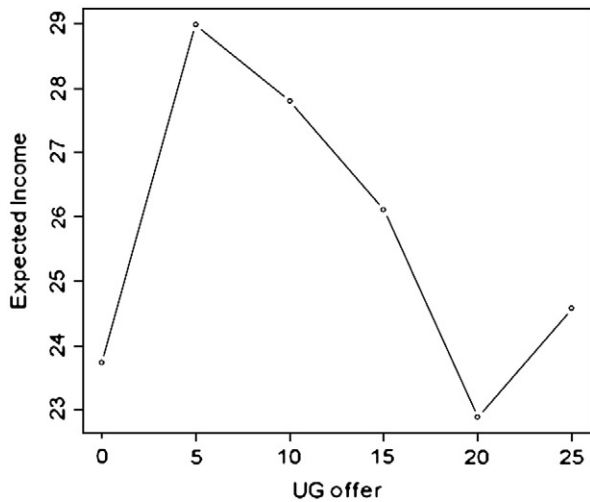


Fig. 4. The expected income (in RMB) of a proposer with respect to her/his offer (in RMB), given the probability distribution of MAO.

4.2. The Estimated Impact of the Fairness Norm in NFCEP

We estimated general conformity to the norm of fairness by deriving the parameter β according to Eq. (1). In the UG, the proposer's surplus benefit equaled the loss of the paired responder, that is, $b = c$. We then rewrote the first equation of Eq. (1) into $U_b = b(1 - 2\beta)$. Obviously, U_b is maximized when $b = 0$ if and only if the marginal utility $dU_b/db \leq 0$. Therefore, $\beta \geq 0.5$ holds for the subject who proposed a fair or hyper-fair offer, and the relevant proportion in the experiment is 96.7%.

If everyone conforms consistently to the fairness norm in the NFCEP as in the experiment, a comparable proportion (96.7%) of the households are expected to choose the strategy of cooperation in forest conservation even without any monitoring pressure, in accordance with the model in which $b < c$. The actual proportion of cooperators may deviate from the expectation when conformity to the social norm varies in different contexts. In general, this prediction coincides with field survey data indicating that only 5 sites out of 387 were found with illegal logging trails when the monitoring intensity was low (Liu et al., unpublished data, 2005).

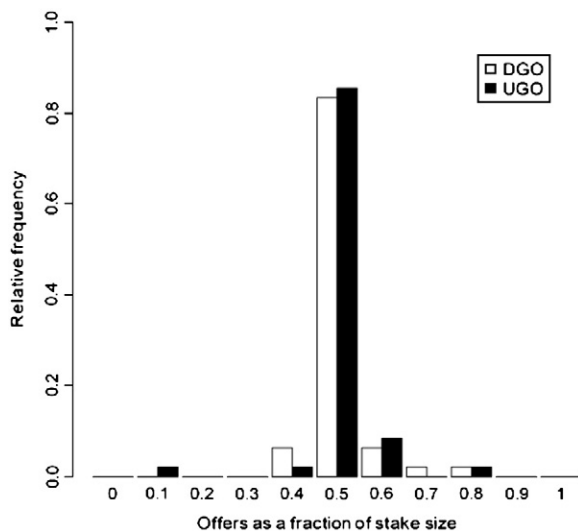


Fig. 5. Offers (in fraction) comparison between Dictator Game and Ultimatum Game. No significant difference was declared, $p = 0.8086$.

5. Discussion

5.1. Linking the Experiment to NFCEP

The UG based experiments were designed to reflect how the subjects conformed to the norm of fairness in a neutral social context, and we did not assume that the social norm must remain strictly constant across different contexts (Bicchieri, 1999). We used context-free game instructions rather than an explicit description of the NFCEP in order to minimize potential experimenter demand effects. Although a specific context may help subjects' understanding in relation to real world experience, like in some other field experiments (e.g., Castillo and Samsel, 2005), at the same time, however, it may distract subjects and allow them to carry over unrealistic expectations to the task and induce experimenter demand effects (Zizzo, 2008). In this specific case, any given cues about NFCEP would likely raise the subjects' concern about the suspected relationship between the experiment and the government, and might consequently lead to a biased response because they knew well the government's expectation.

On the other hand, the sense of justice has deep biological and cultural underpinnings, which probably evolved in the long history, and hence the social norms in the same community must keep some consistency across different contexts (Trivers, 2006). Since NFCEP is only a minor part of the livelihood of the local residents and it has a history of less than a decade, distinct social norms are unlikely to have evolved therein, and the residents plausibly adopted the general traditional norm for decision-making in the unfamiliar task. Thus, it is reasonable to expect consistency to some nontrivial extent from the social norm in the general context to that in NFCEP. In addition, the anonymous one-shot experiment environment provided weaker social bonds than does daily life or the NFCEP, so the experiment result may slightly underestimate rather than overestimate the impact of the social norm in real life.

The post-game interview confirmed the norm of fairness. When asked about the reasons for their choices in the experiment, a majority of the interviewees explicitly attributed it to the preference for fairness, and expressed their antipathy to selfishness and greed (described as "heavy heart" in the local dialect). In contrast, they did not refer to their consideration of the game partners' expectation. As a result, it suggests that the subjects made their decisions in the experiments heuristically based on some simple norm about the outcome itself rather than based on some second-order belief about the partners' expectations which would need communication and more complicated calculation (Charness and Dufwenberg, 2006). But we don't preclude the possible impacts of more complicated social norms in the decision-making in different contexts, and the different social norms, if any, should not be assumed to be mutually exclusive.

Notwithstanding the fact that most responders showed strong reluctance to reject low offers in the experiments, many revealed that they would not cooperate with a "heavy-hearted" person any more. This shows that withholding help by shunning or ostracism is a more popular way to avoid conflicts and retaliation than costly punishment in a long-run repeated game, and that the second-order free rider problem of punishment could thus be avoided (Milinski et al., 2002; Panchanathan and Boyd, 2004). This could also explain the responders' stronger sensitivity against positive deviations from the fair offer, which apparently contradicts the general prediction that negative deviations count more (Loewenstein et al., 1989; Fehr and Schmidt, 1999).

5.2. The Norm of Fairness Evolved with Reputation

The almost ubiquity of the fairness norm across societies with distinct cultures (Henrich et al., 2005) suggests that there might be some biological underpinnings for other-regarding norms in human nature. But the notable variation indicates that ecological and cultural factors probably shape the norms in one way or another. For instance, compared with the results of the preceding cross-cultural experiments

Table 3
Comparison of Ultimatum Game results across villages.

Village	Test	V2			V3			V4		
		Offer	MAO	MXAO	Offer	MAO	MXAO	Offer	MAO	MXAO
V1	EST ^a	0.514	0.496	0.814	0.256	0.740	0.245	0.371	0.599	0.989
	WT ^b	0.631	0.109	0.761	0.205	0.762	0.057	0.350	0.552	0.131
V2	EST	–	–	–	0.899	0.278	0.387	0.992	0.724	0.770
	WT	–	–	–	0.600	0.228	0.136	1.000	0.345	0.967
V3	EST	–	–	–	–	–	–	0.997	0.794	0.238
	WT	–	–	–	–	–	–	1.000	0.762	0.686

^a EST: the *p* values of Epps–Singleton test.

^b WT: the *p* values of Wilcoxon rank sum test (exact test adjusted for ties).

(Henrich et al., 2006), the subjects of Wolong showed relatively stronger preference for fairness in our experiment (Fig. 3). This raises the question of how fairness locally evolved or remained as an important social norm in the small communities of Wolong.

The norm of fairness is unlikely to evolve independently from the experience of NFPC as discussed earlier. Instead, we speculate that the norm might evolve from repeated daily social interactions based on reciprocity (Fehr and Gächter, 2000) and reputation (Nowak et al., 2000). Most of the local residents have lived together in a small community for generations, and may continue to interact for much of their lives, so it is not surprising that cooperation plays an important part in their daily livelihood. Even nowadays, the farm work remains labor intensive (e.g. herding, seeding and harvesting) and most households are too poor to afford the rent of a pasture or some mechanical facilities. They also cooperate voluntarily to build temples and roads. Therefore, they need to build and maintain trustworthiness between each other. The individual decision whether or not to cooperate with another may depend on the co-player's reputation, and fair or hyper-fair offers can help to signal one's generosity and build up one's good reputation (Gintis et al., 2001). In a densely populated community with high viscosity, gossips could amplify the influence of one's reputation (Sommerfeld et al., 2007).

However, for the specific case of Wolong, it is too early to conclude whether the norm of fairness evolved locally or was inherited from the culture and biological nature of the ancestors long before the local community was formed. The answer might be both.

5.3. Sustainable Development and Social Capital

The household-based NFPC has so far worked effectively in Wolong Nature Reserve. However, considering the uncertainty of the future, especially the impact of ongoing economic development, the sustainability of natural resource management in the long run still provides a challenge. On the one hand, the growing exposure to markets may create more opportunities for employment and increased income so that the reliance on fuelwood and other natural resources could be diminished. On the other hand, there is still uncertainty and disagreement over the social impact of the experience with markets. Some evidence hints that the integrity of the market may enhance the norm of fairness (Curven, 2004), whereas the opposite possibility also exists, that the bonding social capital in the local community might decline due to less time and incentive to cooperate with each other (Putnam, 2000).

In Wolong, the subjects with commercial experience with tourism showed a significant higher tendency to be willing to accept lower offers ($n = 12$, $p = 0.0269$), in spite of the small sample. Given the growing economic independence and the potentially declining social capital, the norm of fairness might lose ground, following a similar trend as recognized by Ghimire: “the abandonment of the commune system and influence of modernization have also meant a steady decline in the previous reciprocity system of neighbors helping each other with house building” (Ghimire, 1997).

As a result, not only the economic influence but also the social impact of growing market interactions must be taken into consideration. The sub-optimal scenario cannot be ruled out; i.e. that the limited-size tourism and other business could not bring in sufficient profits to eradicate poverty from the local community, but the social capital would irreversibly decline thereafter and thus cooperation for conservation would break down. For the sake of the sustainability of social–ecological systems, the function and evolution of social capital calls for more attention.

6. Conclusion

The conservation of the giant panda habitat and the ecosystem services in Wolong Nature Reserve has remarkably improved since the implementation of the household-based NFPC, even though the limited material subsidy could not enforce the cooperative commitment of the local residents if they were only self-regarding. This study indicates that the local social norm of fairness may be playing a positive role in facilitating community cooperation in NFPC as the residents are reluctant to benefit from illegal logging at the cost of their neighbors' loss. It therefore calls attention to the need for policy design to take into account the active local social norms. Moreover, in a dynamic view, there is much work to do to unveil how social norms will evolve in changing social–economic contexts, and how this will impact the management of a common natural resource.

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